PassFrame tuition

# Volume & 3D Shapes

Year 4 Pt 2 — Homework ⏱ 45 min

**Worked example — read this through before you start**

**Worked example:** What is the volume of a cuboid measuring 6cm × 4cm × 2cm?
*Method & answer: 6 × 4 × 2 = 48 cm³.*

## Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** What is the volume of a cuboid measuring 5cm × 2cm × 2cm?

**2.** What is the volume of a cube with sides of 2cm?

**3.** Which of these has a volume of 24cm³? A) 4×3×2 B) 6×2×2 C) both A and B D) neither

**4.** How many edges does a cuboid have?

**5.** How many vertices does a triangular prism have?

**6.** What is the volume of a cuboid measuring 7cm × 2cm × 2cm?

**7.** Which 3D shape has 2 flat circular faces and 1 curved surface joining them? A) cylinder B) cone C) sphere D) cube

**8.** What is the volume of a cube with sides of 5cm?

**9.** Select ALL of these that are properties of a cube: A) 6 faces B) all faces equal squares C) 12 edges D) 1 curved face E) 8 vertices

**10.** What is the volume of a cuboid measuring 9cm × 2cm × 1cm?

**11.** How many edges does a square-based pyramid have?

**12.** Which shape has a square base and 4 triangular faces meeting at a point? A) triangular prism B) square-based pyramid C) cube D) cone

**13.** How many faces does a square-based pyramid have?

**14.** What is the volume of a cuboid measuring 6cm × 3cm × 1cm?

**15.** How many vertices does a square-based pyramid have?

## Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** A cuboid has a volume of 72cm³. Its length is 6cm and width is 4cm. What is its height?

**2.** Convert 2000cm³ to litres (1000cm³ = 1 litre).

**3.** Which of these has a volume of MORE than 36cm³? A) 6×3×2 B) 4×3×3 C) both A and B D) neither

**6.** Which net folds to make a triangular prism? A) 2 triangles and 3 rectangles B) 6 squares C) 1 triangle and 3 rectangles D) 2 circles and a rectangle

**7.** A fish tank measures 25cm × 10cm × 20cm. What is its volume in cm³, and in litres?

**8.** Select ALL of these cuboids that have a volume between 40cm³ and 60cm³ (exclusive): A) 6×4×2 B)

**11.** A cube's 6 faces are identical squares, each with an area of 25cm². What is the length of one edge?

**12.** Which shape has the SAME volume as a 5cm×4cm×2cm cuboid? A) a 4cm×5cm×2cm cuboid B) a 4cm×4cm×2cm cuboid C) both A and B D) neither

**13.** A cuboid has dimensions 8cm × 3cm × 4cm. What is its volume?

5×5×2  C) 4×4×3  D) 10×2×1  E) 3×3×6

**4.** A cuboid measures 8cm × 2cm × 2cm. What is its volume?

**5.** Convert 4000cm³ to litres.

**9.** Cuboid A measures 5×2×2cm; cuboid B measures 4×2×3cm. Which has the greater volume?

**10.** A cuboid's volume is 108cm³. Its length is 9cm and height is 4cm. What is its width?

**14.** Convert 2.2 litres to cm³.

**15.** A cuboid measures 6cm × 5cm × 2cm. What is its volume?

## Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** A cuboid's height is 3 times its width of 2cm, and its length is 4cm. What is its volume?

**2.** A cube's edge is 2cm longer than 3cm. What is the cube's volume? A) 27cm³  B) 125cm³  C) 25cm³  D) 15cm³

**3.** A cuboid has the same volume as a cube of side 3cm. If the cuboid's length is 9cm and width is 3cm, what is its height?

**4.** A cuboid's width is a third of its length of 12cm, and its height equals its width. Select ALL of these that are TRUE: A) the width is 4cm  B) the height is 4cm  C) the volume is 192cm³  D) the volume is 144cm³  E) the width is 3cm

**5.** The volume of a cube is the same as a cuboid measuring 8cm×4cm×2cm. What is the length of the cube's edge?

**6.** A tank measures 30cm × 20cm × 15cm and is filled to a depth of 10cm. How many litres of water does it contain?

**7.** A cuboid's length, width and height are in the ratio 3:2:2. If the height is 4cm, what is the volume?

**8.** A cube-shaped box has a volume of 125cm³. What is the length of one edge, and its total surface area?

**9.** A cuboid box has a volume of 90cm³. Its length is 6cm and width is 5cm. What is its height? A) 2cm  B) 3cm  C) 4cm  D) 5cm

**10.** Five identical 2cm cubes are stacked directly on top of each other. What is the total volume, and the height, of the stack?

**11.** Select ALL of these boxes below that could be exactly filled by 36 unit cubes with no gaps: A) 6×3×2  B) 9×2×2  C) 4×3×3  D) 5×4×2  E) 12×3×1

**12.** A cuboid measures 9cm × 4cm × 3cm. It is cut exactly in half, dividing its 3cm height into two equal layers. What are the dimensions of each half?

**13.** A cuboid has a volume of 128cm³. Its length is twice its width, and its height is 4cm. What are its length and width?

**14.** A 16cm×12cm×10cm container, completely full of water, is poured into empty cube-shaped bottles of side 4cm. How many bottles can be completely filled, and how much water is left over?

**15.** A cuboid box has two different face areas: 20cm² and 15cm². The third face has an area of 12cm². What is the volume of the box?